



Angularly resolved polarization microscopy for birefringent materials with Fourier ptychography

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Abstract: Polarization light microscopy is a very popular approach for structural imaging in optics. So far these methods mainly probe the sample at a fixed angle of illumination. They are consequently only sensitive to the polarization properties along the microscope optical axis. This paper presents a novel method to resolve angularly the polarization properties of birefringent materials, by retrieving quantitatively the spatial variation of their index ellipsoids. Since this method is based on Fourier ptychography microscopy the latter properties are retrieved with a spatial super-resolution factor. An adequate formalism for the Fourier ptychography forward model is introduced to cope with angularly resolved polarization properties. The inverse problem is solved using an unsupervised deep neural network approach that is proven efficient thanks to its performing regularization properties together with its automatic differentiation. Simulated results are reported showing the feasibility of the methods.

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1. Introduction

Polarization light microscopy is a popular approach in mineralogy [1], bio-mineralogy [2] and biological studies [3–5]. It is used to reveal local structural properties of the sample [6,7] and, for example assess its internal stress constraints, three-dimensional organization, or its anisotropy [1–3,6,7]. To perform such analysis, one generally uses methods such as polarimetry or cross polarization microscopy [7]. In the usual context of thin planar sample, the latter is generally probed with polarized light under normal incidence. As a result, polarization effects are revealed only along the probing direction, i.e. the optical axis of the experimental setup. New techniques to recover the polarization properties of the sample have been recently demonstrated, using phase imaging methods such as holography or Fourier ptychography [8–10]. Due to the applied forward model, considering the sample only as a Jones matrix, the polarization properties are here again limited to their retrieval along the optical axis. Hence, polarization properties are not resolved angularly, but their 3D orientation is crucial for a more global sample characterization, as it is relevant to its internal structure [11–13]. To overcome this specific limitation, prior knowledge of the material is usually required [11]. Alternatively, three-dimensional polarization properties could be retrieved by probing the sample from different angles as describe in the work of Shin *et al* and Yeh *et al* [12,13]. Hitherto the difficulties rely on how to extract polarization properties from the angular measurements. In these latter works the object is described as the full three dimensional dielectric tensor. They use the angular information to reconstruct the tensor at each voxel of the object, and the inversion methods are based on standard gradient descent approaches.

In this article, we propose a new method to resolve angularly the polarization properties of a thin birefringent sample. The model is restrained to this specific case, as birefringence is common for thin anisotropic materials, and this simplified model of anisotropy will work as a

first proof of concept. The approach is based on Fourier Ptychography Microscopy (FPM) [14] which is modified to take polarization properties into account. Accordingly to FPM principles, the angular information is also used here in order to gain spatial resolution on the reconstructed object. The resulting forward model requires a vectorial formalism and the associated inverse problem is resolved by a Convolutional Neural Network (CNN) following a conditional Deep Image Prior (cDIP) approach. The deep neural network is purposely introduced to solve the inverse problem in order to benefit from its multiple advantages, such as automatic differentiation and more importantly its intrinsic efficient regularization method [15–17]. As the network is based on a cDIP approach, it is unsupervised, i.e. it does not need to learn on a database [18]. The theory and resolution are demonstrated and discussed on a simulation example.

2. Theory

2.1. Forward model

The proposed method in this paper is based on the FPM approach, which is an inverse problem phase imaging method that allows super-resolution imaging on large field of view [14,19,20]. With the latter the object is probed at multiple incidence angles, which is done by the use of a led-matrix [14,19,20]. The field produced after the interaction of the probing light with the sample, namely the exit field, is collected by an objective to produce an image at the sensor plane. The intensity of the field is recorded by a camera. To probe the polarization properties of the object, two linear polarizers are introduced: one before the object to control the polarization of the incident light, and one before the sensor to analyze the polarization of the exit field. The polarizers are mounted on a motor to be able to change their orientations. The modified FPM setup is displayed in Fig. 1. In order to remove any under-determination in the polarizations properties of the object, the latter is probed with three different polarizations (linear at 0, 45, 90°) and the exit field is also analyzed with the same three polarizations. This nine combinations of polarizations/analysis might not be the optimal number of combinations, but it is a safe way to remove polarization-related ambiguities [21].

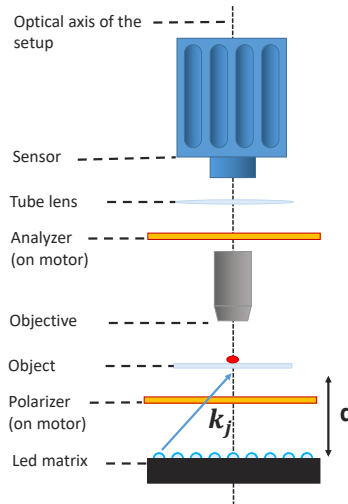


Fig. 1. Modified setup from the original FPM, with the addition of polarizers mounted on controllable motors. The sensor plane is conjugated to the object plane through the objective. The distance between the led matrix and the object is noted d . Each led j light the sample at a different angle, and the direction of the wavevector of this j -th illumination is denoted k_j

Each illumination is considered as a tilted plane wave of wave vector $\mathbf{k} = [k_x, k_y]$. Under thin sample assumption, the tilted incident field is assumed to shift the center of the sample's spectrum by (k_x, k_y) in Fourier domain [14]. The incident light from the j -th illumination propagate at the direction $\mathbf{k}_j$

$$\mathbf{k}_j = \begin{bmatrix} -x_j \\ -y_j \\ d \end{bmatrix}, \quad (1)$$

where x_j and y_j are the coordinates in direct space of the j -th led from the optical axis of the experimental setup, and d is the distance between the led matrix and the object. We define three different polarizations of the light, linearly oriented at an angle θ_n , $\theta_n \in \{0, 45, 90^\circ\}$.

In this article we suppose the sample to be birefringent. Structural organization of birefringent materials presents a symmetry axis (named the optic axis), resulting in two different optical indices. The one along the symmetry axis is known as the extraordinary index n_e and the one orthogonal to the symmetry axis as the ordinary index n_o . These two indices and the orientation of the optic axis are related one another through the index ellipsoid, which completely characterize material birefringence properties. Each axis of the ellipsoid has the length of its index, and the revolution axis $\mathbf{c}$ is oriented the same way as the optic axis. The birefringent object can therefore be model optically by a transmittance factor T and the previously mentioned ellipsoid, known as uniaxial index ellipsoid [7,22]. This description also covers isotropic materials, with $n_e = n_o$, i.e., the ellipsoid being a sphere [7,22]. Retrieving the index ellipsoid of a material is therefore an optical way to characterize its structural properties. The optic axis $\mathbf{c}$ is oriented by its in-plane orientation θ_x and its tilt θ_z (Fig. 2(a)) by the following equation:

$$\mathbf{c} = \begin{bmatrix} \cos(\theta_x) \cos(\theta_z) \\ \sin(\theta_x) \cos(\theta_z) \\ \sin(\theta_z) \end{bmatrix}. \quad (2)$$

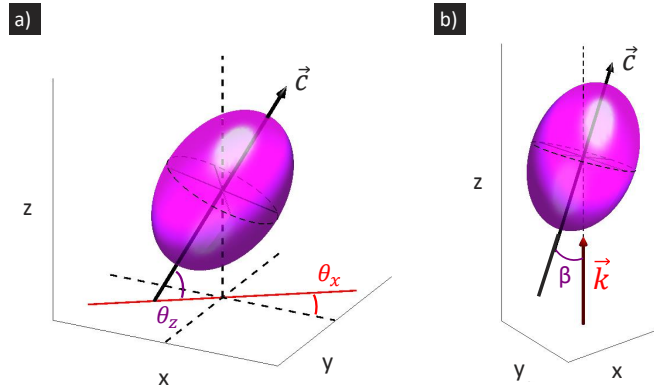


Fig. 2. (a) Index ellipsoid of a birefringent material. Its orientation, i.e., the orientation of its optic axis $\mathbf{c}$, is described by two angles θ_x (azimuthal angle) and θ_z (tilt angle). (b) Angle β between the index ellipsoid and the incident light. The incident light experiments different optical indexes through the matter depending on this angle [7,22].

The interaction between a polarized light and a birefringent object depends on the angle β_j between the propagation direction $\mathbf{k}_j$ of the j -th incident light and the orientation of the object

index ellipsoid $\mathbf{c}$ (depicted in Fig. 2(b)) [6,7,22]. This angle is defined by

$$\cos(\beta_j) = \frac{\mathbf{k}_j}{\|\mathbf{k}_j\|} \cdot \mathbf{c}, \quad (3)$$

and allows us to compute the effective optical index $n_{\text{eff},j}$ with the following expression:

$$n_{\text{eff},j} = \left[\left(\frac{\cos(\beta_j)}{n_o} \right)^2 + \left(\frac{\sin(\beta_j)}{n_e} \right)^2 \right]^{-1/2}. \quad (4)$$

The j -th incident light interacts with an *effective* birefringent object of two indexes n_o and $n_{\text{eff},j}$. The Jones matrix $\mathbf{J}_j$ of this *perceived* object is therefore [23]

$$\mathbf{J}_j(\mathbf{r}) = A(\mathbf{r})\mathbf{R}_\theta \begin{bmatrix} \exp(i\phi_o) & 0 \\ 0 & \exp(i\phi_{\text{eff}}) \end{bmatrix} \mathbf{R}_{-\theta}, \quad (5)$$

with the phase delay $\phi_\bullet$ expressed, depending on the optical index $n_\bullet$, by

$$\phi_\bullet = \frac{2\pi}{\lambda} n_\bullet e, \quad (6)$$

where e is the thickness of the object in each point. This thickness is set to an uniform unitary value, as this method is only sensitive to the phase. In the Jones matrix model, the light-matter interaction is multiplicative, it is therefore possible to follow the FPM equations for the forward model. The incident field $\mathbf{E}_{j,n}^i$ at the j -th led and n -th polarization interact with the effective object $\mathbf{J}_j$ to produce the exit field $\mathbf{E}_{j,n}^e$,

$$\mathbf{E}_{j,n}^e = \mathbf{J}_j \mathbf{E}_{j,n}^i. \quad (7)$$

Following the standard FPM equations [14], the exit field is filtered in frequency by the objective with a filter transfer function P_j in Fourier space, and the sensor measures the intensity of the produced image. In this method we analyze the polarization of the exit field before the intensity measurement, with a linear polarizer l_m , at angle θ_m , $\theta_m \in \{0, 45, 90^\circ\}$. Thus, in our case the sensor measures an intensity $I_{j,m,n}^{\text{meas}}$ for the j -th illumination at the n -th polarization and m -th analysis:

$$I_{j,m,n}^{\text{meas}} = \left\| \mathcal{F}^\dagger \left(l_m P_j \mathcal{F} \left(\mathbf{E}_{j,n}^e \right) \right) \right\|^2. \quad (8)$$

2.2. Inverse problem

The object is represented by a set of optical properties maps ($T; \Phi_o; \Phi_e; \theta_x; \theta_z$). Therefore, the solution S of the problem is the set of maps

$$S = \{T(\mathbf{r}); \Phi_o(\mathbf{r}); \Phi_e(\mathbf{r}); \theta_x(\mathbf{r}); \theta_z(\mathbf{r}), \forall \mathbf{r}\} \quad (9)$$

that minimizes the following criterion $\mathcal{L}$

$$\mathcal{L}(S) = \sum_j \mathcal{L}_j(S) = \sum_j \sum_n \sum_m \left(\sqrt{I_{j,m,n}^{\text{calc}}}(S) - \sqrt{I_{j,m,n}^{\text{meas}}} \right)^2, \quad (10)$$

where $I_{j,m,n}^{\text{calc}}$ is the calculated intensity when the estimated object of solution S is probed by the j -th led at the n -th polarization and m -th analysis. Similarly, $I_{j,m,n}^{\text{meas}}$ is the measured intensity in

the same conditions. The total number of illuminations, polarizations and analysis are noted respectively N_j , N_n and N_m . This criterion is based on the usual ptychographic criterion developed in direct and Fourier ptychography and is well-known for its efficiency in the inversion process and noise robustness for this type of measurements [24]. As FPM probes the object with different spectrum shift, the object is retrieved with a resolution boost [14]. For the sake of clarity, the raw data resolution and retrieved resolutions are noted respectively N_{raw} and N_{rec} .

With the described forward model, the object is reconstructed through projections and non-linear combinations of its optical properties maps. Due to this complexity, a regular iterative gradient-based algorithm like the Ptychographic Iterative Engine (PIE) [14,25,26] would be challenging to directly apply. Indeed, for the latter algorithms, the gradient must be expressed analytically by the derivation of the criterion [27], which could be convoluted in the forward model presented in this paper. Instead, the inverse problem is resolved by the help of a conditional Deep Image Prior (cDIP) Convolutional Neural Network (CNN), based on a UNet architecture, in an optimization process [18,28]. The use of the Deep Neural Network (DNN) solver allows us to benefit from its automatic differentiation, and therefore allows us to use a forward model as complex as needed without having to express its gradient analytically. More importantly, the DNN architecture also offers an intrinsic regularization that helps to solve ill-posed problems and has been proven to perform very well for inverse problem imaging [15–17]. Having a general approach with the DNN to regularize such problem is very interesting since creating an efficient regularization function with its appropriate parameters is a delicate task that must be performed at each modification of the forward problem. This regularization can simply be explained by the way neural networks (NN) works, as explained below.

In inverse problem we want to find the best approximation $\hat{x}$ of the solution x of the general problem

$$Ax \sim y^\delta, \quad (11)$$

where y^δ is our measured data and A a known operator, i.e. our forward model [15]. A neural network can be seen as a function ϕ_Θ parameterized by its neural weights Θ , taking an input z at its entrance. The training of the network will optimize the weights Θ to have the best results according to a loss function. In our case we end up with the following problem to solve:

$$A\phi_\Theta(z) \sim y^\delta. \quad (12)$$

Following the standard approach for inverse problem, we want to minimize the square difference between our output and our measured data by finding the best weights, so we try to minimize the following loss function:

$$\min_{\Theta} \|A\phi_\Theta(z) - y^\delta\|^2. \quad (13)$$

Our approximated solution will therefore be

$$\hat{x} = \phi_\Theta(z). \quad (14)$$

This formulation allows us to directly see that the approximated solution $\hat{x}$ can only be found into the target domain of the function ϕ_Θ . This domain is defined by the architecture of the neural network and its weights. The NN therefore act as a regularization method. This regularization has been demonstrated very efficient for ill-posed problem such as inverse problem in imaging [15–17], with performance sometimes better than the classical Total Value (TV) regularization [16]. The architecture that we choose to use, the UNet, has been specifically praised for its regularization properties in image solving problem [17]. In addition, CNNs are also well-known to be an extremely powerful tool for image processing tasks such as for example denoising [29] or super-resolution [30,31]. Let us note for clarity sake that in our case of conditional Deep Image

Prior (cDIP) our input is our measured data, so

$$z = y^\delta. \quad (15)$$

In order to benefit from deep neural networks without the training on database process, Ulyanov *et al* developed a new approach called Deep Image Prior (DIP) CNN [18]. Deep prior and conditional deep prior CNN have been then successfully used to resolve phase imaging [32], and FPM inverse problem [33], respectively by Wang *et al* in 2020 and Jiang *et al* in 2018. In this article we will name conditional DIP the DIP that use data in input of the NN instead of noise [28,32]. In these latter works, the deep neural networks were unsupervised, i.e. they did not need to be trained and to learn on a set of examples and references. They were built with a physical forward model corresponding to their data in the loss function, that they naturally try to minimize. Therefore, they were able to find a solution to the inverse problem. We adapt the DIP CNN approaches with additional layers to match the direct problem of the polarization FPM (pFPM) and its solution set. The loss function is constructed in two steps. Firstly, the solution S is taken from the output of the CNN and is used by the forward model to generate the calculated intensity images I^{calc} . Secondly, these images are compared to the measured ones I^{meas} by the criterion $\mathcal{L}$ which is described in Eq. (10). The loss is used to generate the gradient that corrects the CNN to minimize the loss. This iterative optimization process stops when the criterion $\mathcal{L}$ stabilizes. The overall workflow is shown on Fig. 3.

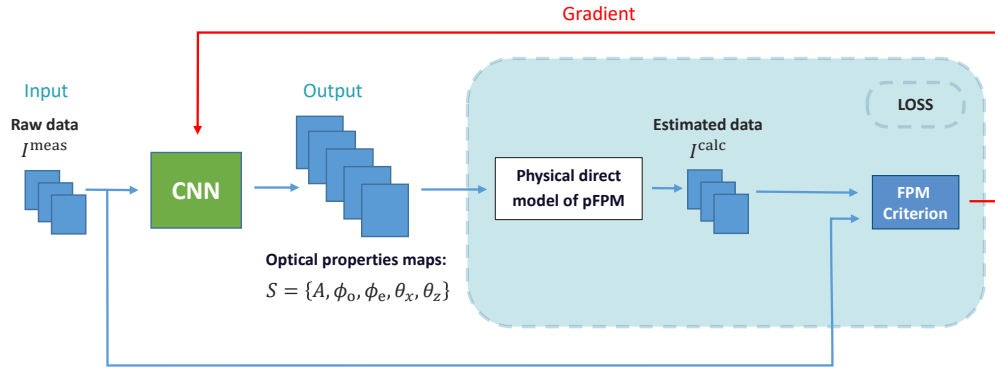


Fig. 3. Inverse problem resolution workflow. A CNN is used to transform the measured data into a solution set adapted to pFPM. The loss function of this network is based on the physical forward model and the pFPM loss function (i.e., criterion). The loss is used to produce a gradient that will correct the CNN. This optimization loop stops when the criterion is low enough.

The CNN which is employed in this work follows the general architecture of a U-Net with skip connections, shown in Fig. 4. This architecture allows to efficiently extract components and the skip connections allows a full use of the data without resolution loss due to the ‘U’ shape [34]. The input layer of the U-Net as the geometry of the input raw data, i.e., $N_{\text{raw}} \times N_{\text{raw}} \times N_j \times N_n \times N_m$. The exit layer meets the solution S geometry $N_{\text{rec}} \times N_{\text{rec}} \times 5$.

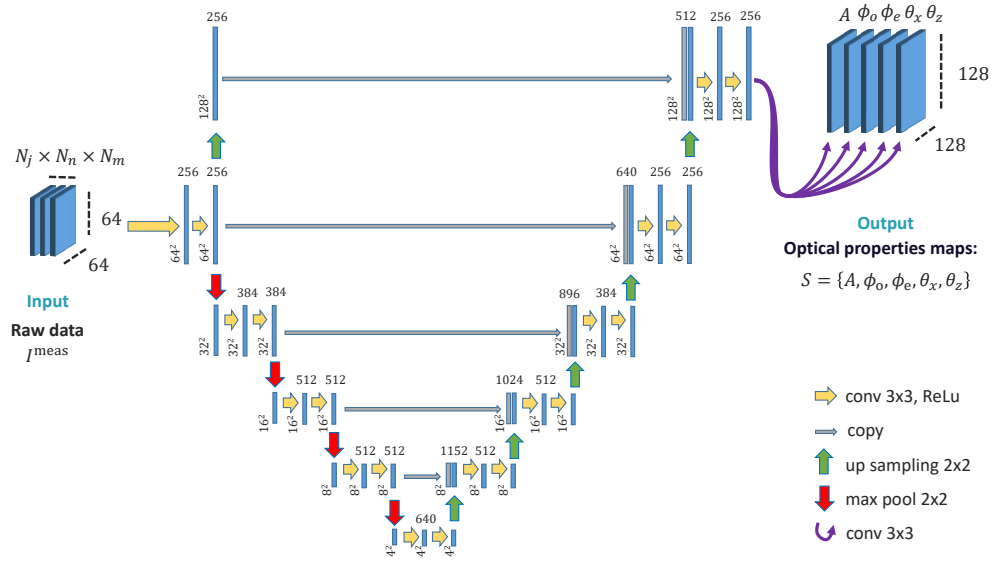


Fig. 4. The used CNN to solve the inverse problem, with for the example $N_{\text{raw}} = 64$ and $N_{\text{rec}} = 128$. The architecture is based on a U-Net with skip connections.

3. Methods and simulation

The feasibility of this method is demonstrated through a simulation, on a birefringent object which is designed to emphasize the capacity of the method to retrieve the three-dimensional orientation of the index ellipsoid. The simulated object has constant thickness and is made of squares in a circular pattern with numbers on each square for the transmittance T . The object is birefringent on all squares, except the one in middle which is isotrop like the background. The optical index along the ordinary axes is uniform on those squares, and uniform on the background. The extraordinary index follows the same behavior. The orientation of the index ellipsoid changes at each square for the in plane orientation θ_x and inside each square for the tilt θ_z . The object optical properties map are summarized in Figs. 5(a)-(f). This object has been conceived for two main reasons:

- (i) If the object is studied with usual polarimetry methods, i.e., probed at normal incidence, the birefringent squares are perceived as having birefringence variation ramp instead of tilt variation ramp.
- (ii) This object shows a lot of diverse configurations and can therefore act as a generic sample. Indeed, the latter present case of both isotropy and birefringence, and the orientations of the birefringent area varies enough to create under-determinations if the problem is solved without combining all the information in the data. For example, the square with the label ‘4’ in transmittance (Fig. 5(a)) has an azimuth orientation θ_x (Fig. 5(d)) aligned with the ‘x’ axis of the referential. Thus, any light with linear polarization at 0° (the first polarization state of the incident light) will perceive it as isotropic. Moreover, if the angular information of the leds are not well used, the object could be seen at each point as a material with its optic axis in the plane ($\theta_z = 0^\circ$), with birefringence variation instead of tilt orientation variation.

The raw data are produced by simulation following the model that is described in part 2.1. The inversion is made with the method explained in part 2.2.

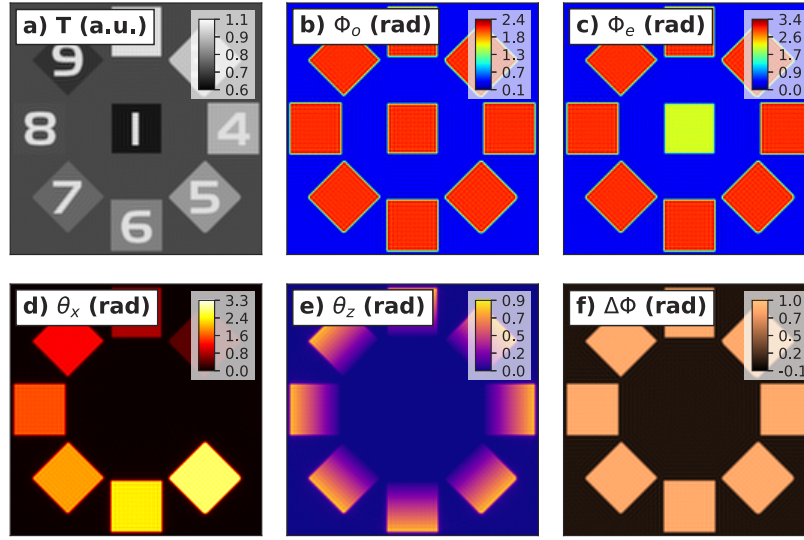


Fig. 5. The reference optical properties maps of the birefringent sample. (a) Transmittance T (u.a.), (b) phase delay in radians ϕ_o (rad) along the ordinary axis, (c) phase delay ϕ_e (rad) along the extraordinary axis, (d) in plane orientation θ_x (rad) of the optic axis, (e) tilt θ_z (rad) of the optic axis, (f) retardance $\Delta\phi$ (rad) between the phase delay of the extraordinary (c) and ordinary axis (b). One can note that the orientation maps of θ_x (d) and θ_z (e) have meaning only when the retardance $\Delta\phi$ (f) is not null.

There is a well-known under-determination coming from Eq. (5), regarding θ_x , concerning birefringent materials. Indeed, the slow and fast axis can be equivalently rotated by $\pi/2$ if their indexes are interchanged. This under-determination should be resolved by Eq. (4), where the latter interchange and rotations are not equivalent. In practice, Eq. (4) have less weight on the calculus of the criterion $\mathcal{L}$ than the Eq. (5) from where rise the under-determination (Eq. (5)). As a result the reconstruction method can be stuck in a local minima. To avoid this local minima, one can initialize the solution S to a better estimation, instead of the usual random initialization. To do so, the inversion is run twice in different solution spaces, the first inversion behaving as an initialization for the second inversion:

- (i) The first time, the network work with two different maps of θ_x : one for each of the two equations Eq. (4) and Eq. (5). This releases a constraint for the network and allows it to retrieve a better solution for Eq. (4) and Eq. (5) separately. Ideally, the two solutions should be the same, as they represent the same property. But, due to the geometry of the problem previously explained, an under-determination could rise. This first reconstruction stops when the criterion is stable.
- (ii) After that, an other inversion is launched with the previously achieved neural weight (so with an initialization of the neural network by the previous solution). But the solution space is set to the regular five maps solution set, using for θ_x the initialization given by the network for Eq. (4).

This inversion in two step allows the network to avoid the local minima due to the previously described high-impact under-determination.

4. Results

The loss of the initialisation inversion and of the inversion are respectively displayed on Figs. 6(a) and (b). The general behavior shows a decrease of loss through the epochs, and a descent and stabilization toward a minima. The figure also shows some very localized loss jump (increase and decrease) in an almost periodic fashion, but they do not seem to interfere with the convergence of the system. The loss for the initialization reaches a plateau around epoch 20000 over 40000 and is mostly stable, meaning that the solution has attained a minima and will not move from it. The second inversion allows the system to jump out of this local minima and converges toward a far

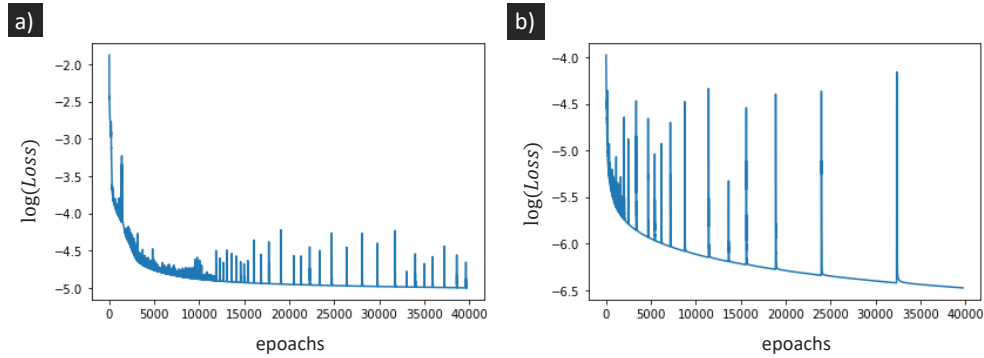


Fig. 6. (a) The loss in log scale in function of the epochs (iterations) for the first inversion, i.e., the initialization. (b) The loss in log scale in function of the epochs for the second inversion, starting from the results of the first inversion.

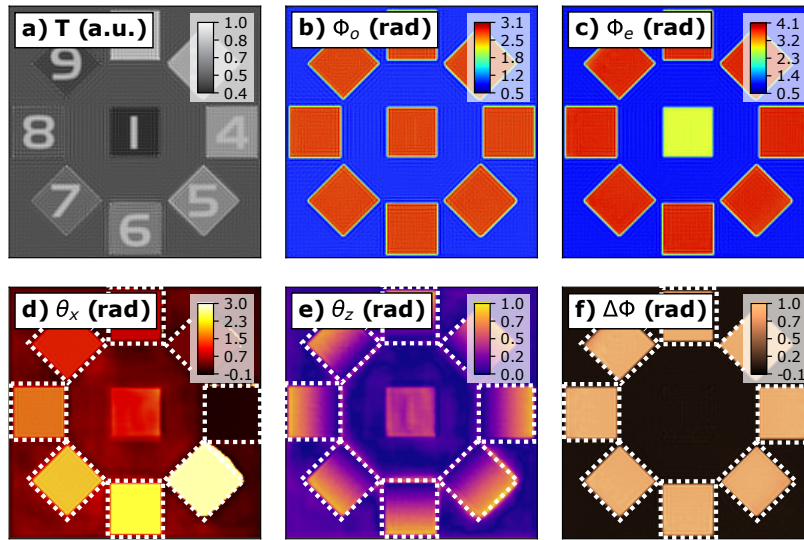


Fig. 7. The reconstructed optical properties maps of the birefringent sample. (a) Transmittance T (u.a.), (b) phase delay in radians ϕ_o (rad) along the ordinary axis, (c) phase delay ϕ_e (rad) along the extraordinary axis, (d) in plane orientation θ_x (rad) of the optic axis, (e) tilt θ_z (rad) of the optic axis, (f) retardance $\Delta\phi$ (rad) between the phase delay of the extraordinary (c) and ordinary axis (b). One can note that the orientation maps of θ_x (d) and θ_z (e) have meaning only when the retardance $\Delta\phi$ (f) is not null. Therefore the reconstructed properties are quasi identical to the reference.

lower loss. The solution achieved at the last epoch is depicted on Fig. 7. The transmittance T (Fig. 7(a)) is well retrieved, both in general shape and value. The reconstruction displays some faint high frequency artifacts, in a grid fashion, but they do not limit the interpretation of the results due to their small amplitude. We also note that these frequencies are present in the reference, but at much lower amplitude. The phase delay along the ordinary axis ϕ_o (Fig. 7(b)) and extraordinary ϕ_e (Fig. 7(c)) are retrieved identical to the reference (respectively Fig. 5(b) and Fig. 5(c)), with the usual global phase shift inherent of phase measurement method without reference. But one can notice that their respective variations are well preserved: the retardance $\Delta\phi = \phi_e - \phi_o$ shown in Fig. 7(f) is strongly similar to the reference Fig. 5(f). The maps of θ_x (Fig. 5(d)) and θ_z (Fig. 5(e)) only have meaning on point where the object is not isotrop, i.e., the retardance $\Delta\phi \neq 0$. The retardance map is displayed on Fig. 7(f), and the area of where the value are not null are circled with a dash line on this map and on θ_x Fig. 7(d) and θ_z Fig. 7(e) maps. On these area, the angles θ_x and θ_z are perfectly reconstructed. The map of θ_z shows a precisely defined gradient of values identical to the reference (Fig. 5(e)). The inversion method successfully retrieves a solution almost identical to the reference, with a low amplitude high frequency small artifact on the transmittance but a great accuracy otherwise.

5. Conclusion

In this work, we have proposed a new method based on FPM which not only opens FPM to birefringent materials but allows to reconstruct their polarization properties with angular resolution. This method gives access to the information of the index ellipsoid of the object locally, with the resolution boost and wide field of view of FPM. For this method we developed an adequate vectorial formalism for polarization FPM, which needed a powerful solver for the resolution of its associated inverse problem. Regarding of the complexity of the forward problem, the inverse problem is solved by a CNN U-net with a loss function based on the physical modeling of the direct problem, using only the ‘experimentally measured’ data to inverse, in an optimization process. The method and its solver are demonstrated by simulations and satisfying reconstructed results were reported.

The resolution of the inverse problem expressed under a deep neural network model will allow us in future work to develop further the forward model and account for more complex consideration such as precise noise model and other fine experimental consideration without having to modify our resolution method. More importantly it can open the path to direct resolution of the inverse problem by trained neural network on a experimentally acquired database, for real time imaging [17]. Based on FPM, this method makes it possible to assess quantitatively angularly resolved polarization information on large field of view at high numerical aperture. This could be relevant for biological imaging, such as for example brain tumor detection [35], collagen orientation for cartilage, bone or skin defect analysis [3–5].

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are available from the authors upon request.

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